MASTER IN ASTROPHYSICS
AND SPACE SCIENCE

TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

Introduction to Numerical Methods for Astrophysics

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Overview

Overview

Fluid Elements

- *Fluids:* things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements:* region with definable local variables, fulfilling certain conditions
  - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

  - large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

  - in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

- *Eulerian View:* cells of small volumes in fixed grid, flow is defined for all t at position r with rate of change $\partial/\partial t$ for any measurable quantity Q
- *Lagrangian View:* follow particular fluid elements, comoving reference system with flow evaluated in each one via D/Dt as the rate of change
- *Translating Relation:* add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + u \cdot \nabla Q$$

Continuous Variables

- *Primitive Fields:*
  - velocity $\boldsymbol{u}$ *(vector)*
  - density $\boldsymbol{\rho}$ *(scalar)*
  - pressure $\boldsymbol{p}$ *(scalar)*
  - temperature $\boldsymbol{T}$ *(scalar)*
- *Conserved Quantities:*
  - mass $\boldsymbol{M}$ *(scalar)*
  - energy $\boldsymbol{E}$ *(scalar)*
  - momentum $\boldsymbol{P}$ *(vector)*
- *Derived Values:*
  - viscosity $\boldsymbol{\eta}$ *(scalar)*
  - vorticity $\boldsymbol{\omega}$ *(vector)*

Overview

Dimensionless Numbers

- *Reynolds Number:* ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{uL}{\nu}$$

- $R < 1000$**:** Laminar Flow (reversible and ordered stream)
  - $R < 1$**:** Stokes Regime (without any separation)
  - $10 \lesssim R \lesssim 40$**:** Separation Zone (stable and stationary vortices)
- $R > 1000$**:** Turbulent Flow (stochastic and chaotic stream)
  - $40 \lesssim R \lesssim 4000$**:** Transitional Kármán Street (vortex shedding)
  - $R > 4000$**:** Kolmogorov Energy Cascade (fully developed phase)

Dimensionless Numbers

- *Mach Number:* ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

  - $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
  - $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
  - $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)
  - $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

Bulk Transport

- *Continuity Equation:* density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

- *Linear Advection:* velocity field $\boldsymbol{v}$ is free of divergence for incompressible fluids

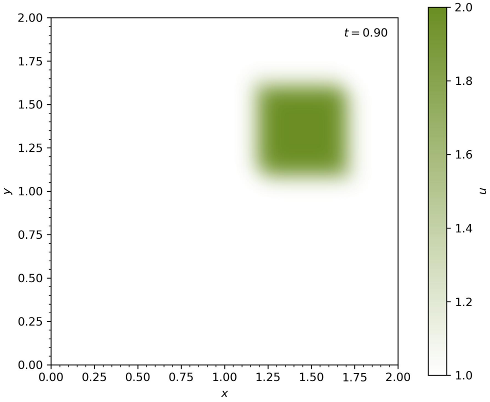
$$\frac{\partial u}{\partial t} + (v \cdot \nabla) u = 0$$

- *Nonlinear Advection:* velocity field $\boldsymbol{u}$ is carried by itself instead of fixed background flow

$$\frac{\partial u}{\partial t} + (u \cdot \nabla) u = 0$$

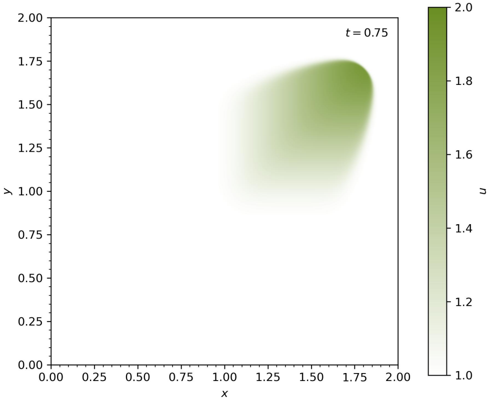
Linear Advection

- *Initialization:*
  - square wave
- *Boundaries:*
  - *Lower:* constant value
  - *Upper:* unconstrained
- *Observation:*
  - fixed profile
  - linear translation



Nonlinear Advection

- *Initialization:*
  - square wave
- *Boundaries:*
  - *Lower:* constant value
  - *Upper:* unconstrained
- *Observation:*
  - shock formation
  - rarefaction tail



Stochastic Transport

- *Diffusion:* kinematic viscosity ν describes spread due to random motion

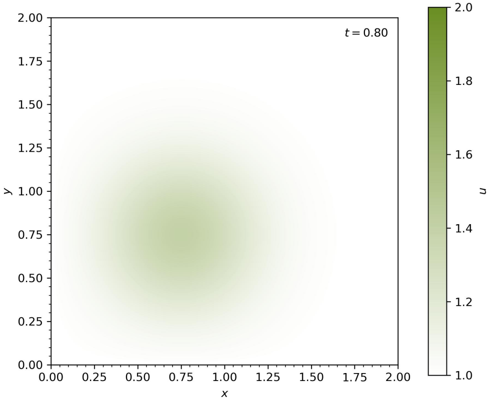
$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation:* combines shock generation with counteracting diffusive term

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = \nu \nabla^2 u$$

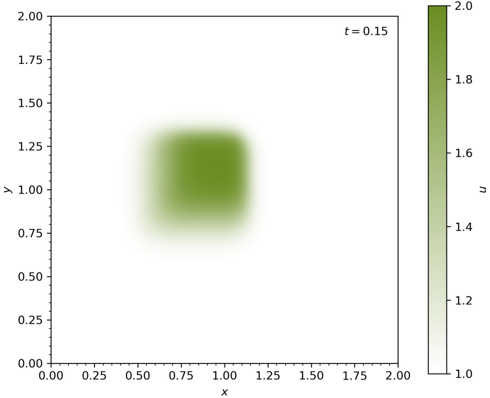
Diffusion

- *Initialization:*
  - square wave
- *Boundaries:*
  - *Both:* constant value
- *Observation:*
  - stationary decay
  - edge smoothing
  - spreading out



Bateman-Burgers Equation

- *Initialization:*
  - sawtooth wave
- *Boundaries:*
  - *Both:* periodic
- *Observation:*
  - nonlinear steepening
  - viscous smoothing
  - dissipating translation



Equilibrium Search

- *Laplace Equation:* maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

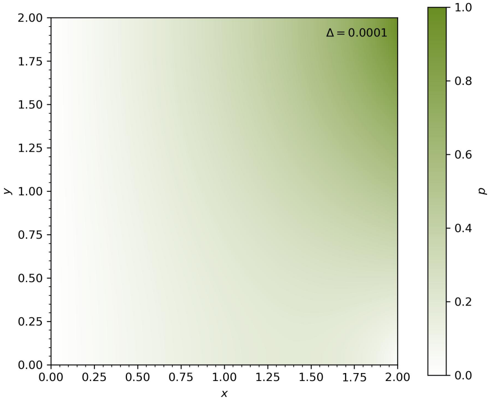
- *Poisson Equation:* extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

  - *Gravity:* $\nabla^2 \Phi = 4\pi G \rho$
  - *Electrostatics:* $\nabla^2 V = -\rho / \varepsilon$
  - *Pressure:* $\nabla^2 p = \rho \nabla \cdot \big((u \cdot \nabla) u \big)$

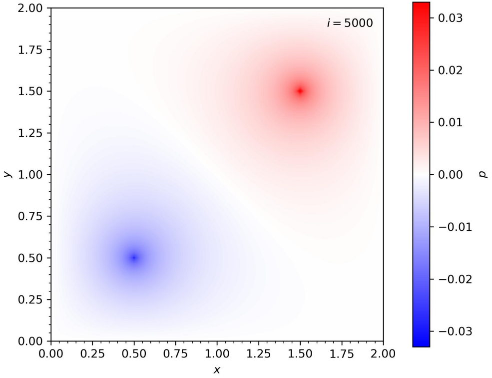
Laplace Equation

- *Initialization:*
  - empty field
- *Boundaries:*
  - *Left:* fixed zero
  - *Right:* linear slope
  - *Top & Bottom: no gradient*
- *Observation:*
  - iterative diffusion
  - smooth relaxation



Poisson Equation

- *Initialization:*
  - empty field
  - positive source
  - negative sink
- *Boundaries:*
  - *All:* fixed zero
- *Observation:*
  - dipole field potential
  - converging relaxation



Navier-Stokes Equations

- *Momentum Equation:* ensure $\nabla \cdot \boldsymbol{u} = 0$ by including $\nabla \boldsymbol{p}$ to instantaneously cancel out any divergence

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\frac{1}{\rho}\nabla p + \nu \nabla^2 u$$

- *Pressure Poisson:* require $\boldsymbol{p}$ to have relaxed to its equilibrium for all $\boldsymbol{t}$ before projecting

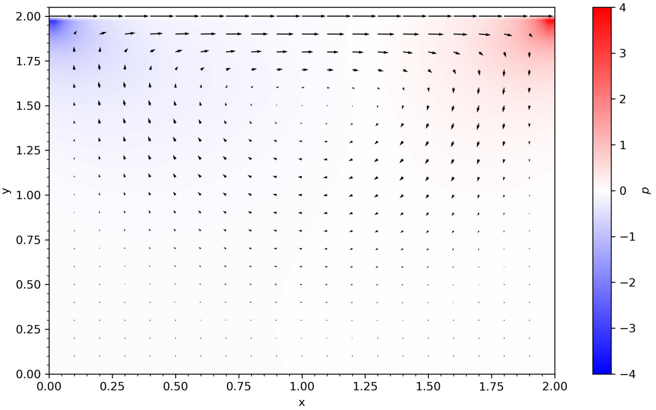
$$\nabla^2 p = \rho \nabla \cdot \big((u \cdot \nabla)u\big)$$

- fulfill at each step to accurately evolve initial configuration under given boundary conditions

Overview

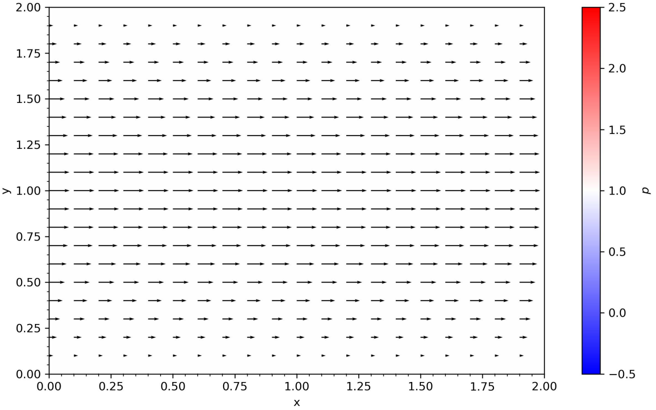
Lid Driven Cavity Flow

- *Initialization:*
  - vanishing velocity
  - zero pressure
- *Boundaries:*
  - *Top:* $\boldsymbol{u} = 1$ & $\nabla p = 0$
  - *Bottom:* $\boldsymbol{u} = 0$ & $p = 0$
  - *Left & Right:* $\boldsymbol{u} = 0$ & $\nabla p = 0$
- *Observation:*
  - vortex formation
  - pressure gradient



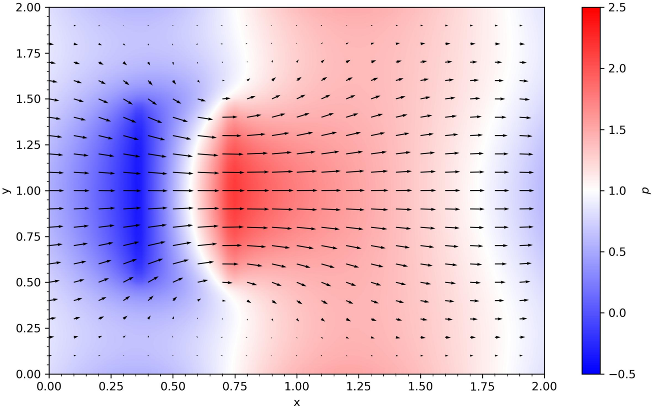
Pressure Driven Channel Flow

- *Initialization:*
  - vanishing velocity
  - unity pressure
  - *Poiseuille:* constant source
- *Boundaries:*
  - *Top & Bottom:* $\boldsymbol{u} = 0$ & $\nabla p = 0$
  - *Left & Right:* periodic
- *Observation:*
  - curved velocity profile
  - boundary layer formation



Locally Forced Channel Flow

- *Initialization:*
  - vanishing velocity
  - unity pressure
  - rectangular source
- *Boundaries:*
  - *Top & Bottom:* $u = 0$ & $\nabla p = 0$
  - *Left & Right:* periodic
- *Observation:*
  - central jet formation
  - recirculation eddies



Overview

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
  - easy implementation
  - computationally efficient
  - highly accurate for simple problems
- Drawbacks:
  - struggles with complex geometries
  - strong numerical diffusion at low orders
  - inflexible rectilinear grid with fixed resolution
  - mass and energy can leak due to numerical errors

Integration Techniques

- *Spatial:*
  - *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
  - *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
    - used to model transport with defined direction
  - *Central Difference:* considers both directions, more precise but can lead to oscillations
    - used to model symmetric dispersion processes
  - *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions
- *Temporal:*
  - *Backward Euler:* looks to future to calculate next step implicitly by inversion, extremely stable
  - *Forward Euler:* looks to past to find next step explicitly, simple but unstable for large intervals
  - *Leapfrog:* reaches back to previous step instead of current one, symplectic and invertible

Integration Techniques

- *Advanced:*
  - *Lax-Wendroff:* take intermediate step, equivalent to series expansion to second order
  - *Runge-Kutta:* take trial steps to estimate curvature, then make informed jump
  - *Predictor-Corrector:* take roughly predicted step, then recalculate with new information
  - *Leapfrog:* preserve phasespace volume and total energy, enable backward reconstruction
- *Courant-Friedrichs-Lewy Condition:* $\frac{v\Delta t}{\Delta x} \leq 1$
- *Pure Diffusion Condition:* $\frac{2\nu\Delta t}{\Delta x^2} \leq 1$

Discretized Formulation

- *Convection:* $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$
- *Diffusion:* $\frac{\partial u}{\partial t} = \nu\frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} = \nu\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$
- *Bateman-Burgers Equation:* $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = \nu\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$
- *Laplace Equation:* $\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \longrightarrow \frac{p_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} = 0$

Overview

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t}\int_V Q\,dV + \oint_S (F\cdot n)\,dS = \int_V \Sigma\,dV$$

- Advantages:
  - inherently strictly conservative for total mass and energy
  - unstructured grid handles complex geometries well
  - easily deals with shock discontinuities
- Drawbacks:
  - mathematically complex at higher order accuracy

Overview

Conceptual Overview

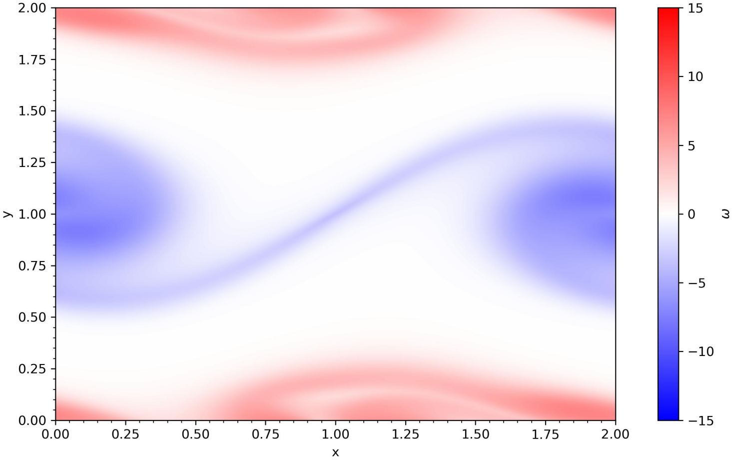
- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
  - highly mathematically robust
  - precise gradient calculation at boundaries
  - handles extremely complex mesh geometries well
- Drawbacks:
  - computationally expensive and memory intensive
  - bad stability for fast turbulent flows

Overview

Setup

- *Initialization:*
  - oppositely directed horizontal velocity layers
  - thin shear boundary modeled with $\tanh(y)$ for smooth transition
  - vertical perturbation using $\sin(x)$ to trigger instability
  - scalar curl $\omega = \partial v/\partial x - \partial u/\partial y$ as visualization
- *Boundaries:* fully periodic for effectively toroidal domain
- *Observation:*
  - shear layer rolls up into distinct spiral structures
  - vortices grow into rotating billows
  - mixing occurs and creates turbulence

Visualization

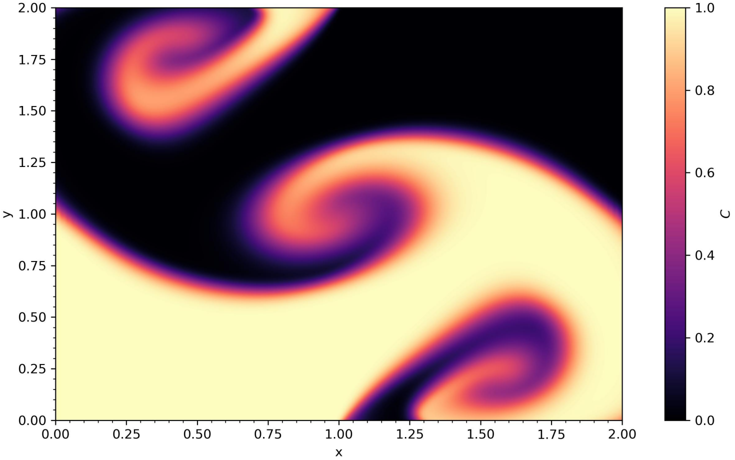


Overview

Setup

- *Initialization:*
  - fluid stratification with vertically resting layers
  - thin interface modeled with $\tanh(x)$ for smooth transition
  - horizontally perturbed using $\sin(y)$ to trigger instability
  - concentration C to calculate coupled buoyancy forces F and as visualization
  - *Boussinesq:* density approximated to be the same in both fluids
- *Boundaries:*
  - *Top & Bottom:* periodic for an infinitely repeating domain
  - *Left & Right:* fixed values to contain the fluids inside plot
- *Observation:*
  - fingers extend from intability and form mushroom caps
  - *Kelvin-Helmholtz Instabilities:* spiraling vortices from friction lead to turbulent mixing

Visualization



Overview

Outlook

- *Frameworks:*
  - *Smoothed Particle Hydrodynamics:* mesh free interactions based on kernel weighting
  - *Lattice Boltzmann Method:* cells simulate statistical particle collisions and streaming
  - *Adaptive Mesh Refinement:* grid resolution increases where higher precision is required
- *Regimes:*
  - *Plasma:* use magnetohydrodynamics
  - *Relativity:* use relativistic hydrodynamics
  - *Degeneracy:* need quantum effects
  - *Collisionless:* need initial distributions
- *Applications:*
  - stellar evolution
  - accretion disks
  - cosmic filaments

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*[1] 2022
- P. Mocz, *Medium*[2] 2020-2025

Resources on GitHub: here